Clinical Guideline for:
Postpartum Haemorrhage
Eastern Maternity Services

Summary

This guideline outlines the management of postpartum haemorrhage in all care settings.

Initial Management / Minor PPH in all Birth Settings

- Call for help – Alert coordinating midwife and obstetric registrar (if in acute hospital)
- Lie flat
- Facial oxygen
- Palpate uterus and rub up a contraction
- Consider bi-manual compression if bleeding not controlled
- Give second dose of IM oxytocic (syntometrine 1 amp IM (contains oxytocin 5iu and ergometrine 500mcg) or Oxytocin (Syntocinon) 10iu IM) and consider additional uterotonics where available –see table below
- Establish IV access – (2x 14 or 16 gauge) and take blood for FBC, Coag, G&S, U&Es
- Commence 1 litre of Hartmanns and give 1g IV Tranexamic Acid
- Insert urethral catheter and empty the bladder
- Ensure placenta is complete and examine for perineal trauma
- Monitor maternal condition – MEWS observations every 5 minutes until stabilised.
- Commence fluid balance in chart
- Document timings and interventions on Emergency Narrator Tab

Drugs for an Atonic Uterus

- Second dose of IM oxytocic (syntometrine 1 amp IM(contains oxytocin 5iu and ergometrine 500mcg) or Oxytocin (Syntocinon) 10iu IM) **or**
- Ergometrine 500mcg intramuscularly or intravenously slowly (N.B. not in patients with hypertensive disease)
- Commence an infusion of 500ml normal saline with 40 units Oxytocin (Syntocinon) infused over 4 hours at 125mls/hr
- Haemabate (prostaglandin F2α) 250micrograms IM repeated every 15 minutes up to a maximum of 8 doses (avoid in asthmatics unless absolutely necessary due to the risk of bronchospasm)
- Misoprostol 800mcgs SL/PR
- Consider bimanual compression until drugs are active

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1.0 DEFINITION

1.1 RCOG describes PPH as minor (500-1000mls) or major (>1000mls). Major can be divided to moderate (1000-2000ml) or severe/massive (>2000ml).

1.2 For clarity and audit purposes our definitions are:

Description	EBL	Response	Management
Minor	500-1000mls		Follow flow chart page 5
Moderate	1500-2000mls >1000mls	Responding to medical treatment Not responding to med treatment	Follow flow chart page 5&6
Major/Massive	>2000mls or		Follow flow chart page 5&6 and instigate massive blood loss protocol
	>1500mls and	Not responding to med treatment	

1.3 Significant PPH may occur rapidly. Blood loss is difficult to accurately assess and is often underestimated. The normal circulatory volume in a pregnant woman is 100mls/kg and hence 1000mls of blood loss in a 50kg women will impact more significantly than in a 100kg woman. Close observation including pulse, blood pressure, respirations (using a MEWS chart) and on-going bleeding is vital. If bleeding does not settle after delivery, appears excessive (>1000mls) or the patient shows signs of compromise (pallor, tachycardia, hypotension) then major blood loss must be presumed and managed as detailed on flow chart on page 5.

1.4 Rarely, women may have concealed haemorrhage (internal bleeding after a caesarean section or a broad ligament haematoma) and present with symptoms and signs of hypovolaemia.

2.0 RISK FACTORS AND ODDS RATIOS FOR PPH (RCOG)

2.1 Antenatal factors associated with a substantial increase in the incidence of PPH include:

- Suspected or proven placental abruption x13.
- Known placenta praevia x12.
- Multiple pregnancy x5.
- Pre-eclampsia/gestational hypertension x4.

Women with any of the above factors should be advised to deliver in a consultant led maternity unit.

2.2 Antenatal factors associated with a significant (although smaller) increase in the incidence of PPH

- Previous PPH x3.
- Asian ethnicity x2.
- Obesity (BMI>35) x2.
- Anaemia (<9g/dl) x2.
- Age (>40 years if primip) x 1.4
- Women with pre-existing bleeding disorders
- Women taking anticoagulants

The above factors should be taken into account when discussing place of delivery.

2.3 Risk factors that may become apparent during labour and delivery:

- Delivery by emergency caesarean section x4.
- Delivery by elective caesarean section x2.
- Induction of labour x2.
- Retained placenta x5.
- Operative vaginal delivery x2.
- Prolonged labour (>12 hours) x2.
- Big baby (>4kg) x2.
- Pyrexia in labour x2.

Any of the above factors should prompt extra vigilance among clinical staff who should anticipate potential major blood loss.

3.0 PREVENTION OF PPH

3.1 Active management of the third stage of labour lowers maternal blood loss and reduces risk of PPH.

3.2 Consider the use of Tranexamic Acid 0.5mg-1g IV, in addition to oxytocin at Caesarean section to reduce blood loss in women at high risk of PPH.

3.3 Prophylactic Uterotonics should be offered:

- Syntometrine 1 amp IM (contains oxytocin 5iu and ergometrine 500mcg) is advised in the absence of hypertension as it reduces the risk of minor PPH but does increase vomiting – See Clinical Guideline for Care of Women in Labour in All Care Settings.

  Onset of action – 2mins 30 secs for oxytocin and 7 mins for ergometrine.
- Oxytocin (Syntocinon) 10 iu IM is advised in vaginal deliveries in hypertensive women
- Carbetocin 1 amp IV (100mcg) or Oxytocin (Syntocinon) 5iu IV followed by an Oxytocin (Syntocinon) infusion should be used for women delivering by Caesarean section. Onset of action - immediate
- Consider additional IV infusion of 40iu of Oxytocin (Syntocinon) in 500mls of normal saline at 125mls/hr if multiple risk factors. **Do not draw up oxytocin until baby has delivered.**
- **Ensure IM Uterotonics are given in the deltoid (arm) rather than the leg in women with BMI>40 or those women where it appears the drug will not reach the muscle when a green needle is used.**

4.0 CAUSES OF PPH – REMEMBER THE 4 'T'S:

- **T**one e.g. Atonic uterus (70%)
- **T**issue e.g. Retained products of conception (10%)
- **T**rauma e.g. Genital trauma (20%)
- **T**hrombin e.g. DIC (1%)

5.0 MANAGEMENT OF POSTPARTUM HAEMORRHAGE

5.1 The management of major obstetric haemorrhage essentially follows the same principals whatever the cause of blood loss.

- A and B – assess airway and breathing. Provide oxygen in major/massive PPH
- C – evaluate circulation – restore and maintain circulating volume
- Basic measures consist of monitoring and readiness for resuscitation
- The source of bleeding identified and dealt with promptly

5.2 IF IN A STAND-ALONE BIRTH CENTRE OR AT HOME

- Call for help
- Midwives on call
- 999 Ambulance
- Minor Injuries Unit (MIU) staff/doctors (to insert cannula), but do not let this delay transfer
- Inform RD&E Labour ward of imminent transfer

The most important priority is to get the woman to an Acute Maternity Hospital ASAP - see Clinical guideline for maternal transfer by ambulance.

Initial Management

Initial Management/ Minor PPH In all birth place settings →

- Call for help – Alert coordinating midwife and obstetric registrar (if in acute hospital)
- Lie flat
- Facial oxygen
- Palpate uterus and rub up a contraction
- Exclude uterine inversion
- Consider bi-manual compression if bleeding not controlled
- Give second dose of IM oxytocic (syntometrine I amp IM or Oxytocin (Syntocinon) 10iu IM) and consider additional uterotonics where available –see table below
- Establish IV access –(1x 14 or 16 gauge) and take blood for FBC, Coag, G&S, U&Es
- Commence 1 litre of Hartmanns
- Insert urethral catheter and empty the bladder
- Ensure placenta is complete and examine for perineal trauma
- Monitor maternal condition – MEWS observations every 5 minutes until stabilised.
- Commence fluid balance chart
- Document timings and interventions

Drugs for an Atonic Uterus

- Second dose of IM oxytocic (syntometrine 1 amp IM or Oxytocin (Syntocinon) 10iu IM) **or**
- Ergometrine 500mcg intramuscularly or intravenously slowly (N.B. **not in patients with hypertensive disease**)
- Commence an infusion of 500ml normal saline with 40 units Oxytocin (Syntocinon) infused over 4 hours at 125mls/hr.
- Hemabate (prostaglandin F2α) 250micrograms IM repeated every 15 minutes up to a maximum of 8 doses (avoid in asthmatics unless absolutely necessary due to the risk of bronchospasm) – Onset of action is 5 mins- peak effect 15 mins
- Misoprostol 800mcgs SL/PR. Onset of action SL is 10 mins whereas PR is 100mins
- Consider bimanual compression until drugs are active.

Call for Additional Help

- Dial 2222 and state 'Obstetric Emergency'
- Alert theatre staff
- Give high flow O_2 via face mask
- Ensure PPH trolley collected
- Assign one midwife for observations and record keeping on the Emergency Narrator tab
- Inform Obstetric Consultant

Restore Circulating Volume

- Insert 2 large bore IV Cannulae if not already sited
- Ensure woman flat and kept warm (Bair Hugger)
- Take blood for Xmatch (6 units initially), FBC, U&E's and Coag
- Rapid replacement of fluid crystalloid via fluid warmer and pressure infuser – avoid excessive crystalloid infusion
- Transfuse Xmatched blood ASAP
- 2 units of O negative blood are kept on LW fridge – if required/need >3units Xmatched blood or in situations of massive uncontrolled haemorrhage– instigate the major obstetric haemorrhage protocol

Correct Coagulopathy

- Give Tranexamic acid 1g IV
- Repeat FBC and Coag after 4 units blood to detect haemostaic failure
- 1 unit FFP for every 1 unit blood
- 2 units cryoprecipitate if fibrinogen <2g/.dl
- If platelet count <50 , give platelets – involved haematology
- Consider use of TEG to guide management of coagulopathy

Monitor

- Vital signs, Pulse, BP, Resps – 5 minutely until continuous monitoring can be arranged as dictated by woman's condition – Record on MEWS chart. Monitor temperature every 15 minutes,
- Maternal oxygen saturation
- Hourly urine output
- Strict Fluid balance
- Monitor and calculate blood loss

Medical/Surgical Management

If EBL>2000mls, O -ve blood used or pt required>3u Xmatch blood – massive blood loss and activate massive blood loss protocol

- Consultant Obstetrician should be present.
- Do not delay transfer to theatre for EUA/LAPAROTOMY
- Manage uterine atony with bimanual compression and drugs (table 1)
- Check for retained placenta/ cervical trauma
- Use cell salvage as required – See Clinical Guideline for use of intraoperative cell salvage (ICS) in obstetrics
- Intracavity balloon –see point 7.1
- B Lynch haemostatic suture – point 7.2 figure 1
- Celox PPH – point 7.3
- Interventional radiology – see point 7.4
- Uterine/internal artery ligation/TAH – call gynaecologist if available
- Transfer to ITU

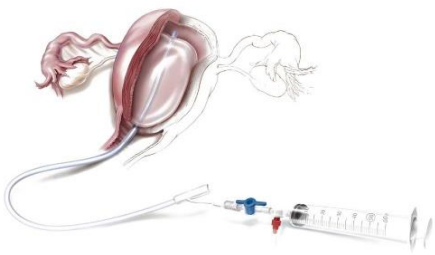
6.0 MASSIVE BLOOD LOSS PROTOCOL

- If EBL>2000mls (or >1500mls not responding to medical intervention), O -ve blood used or the patient has required >3u Xmatch blood, the major obstetric haemorrhage protocol should be instigated.
- To activate the major obstetric haemorrhage protocol – Ring 2222 and state 'I am activating the major obstetric haemorrhage protocol'
- Give your extension number and blood transfusion will call you back.
- A blood pack will be issued containing 4 units of blood and 4 units FFP. A runner will need to be nominated to go to the lab to collect the blood products.
- Once patient is stabilised ensure lab is alerted to 'Stand Down Protocol'

7.0 ADDITIONAL/SURGICAL MANAGEMENT

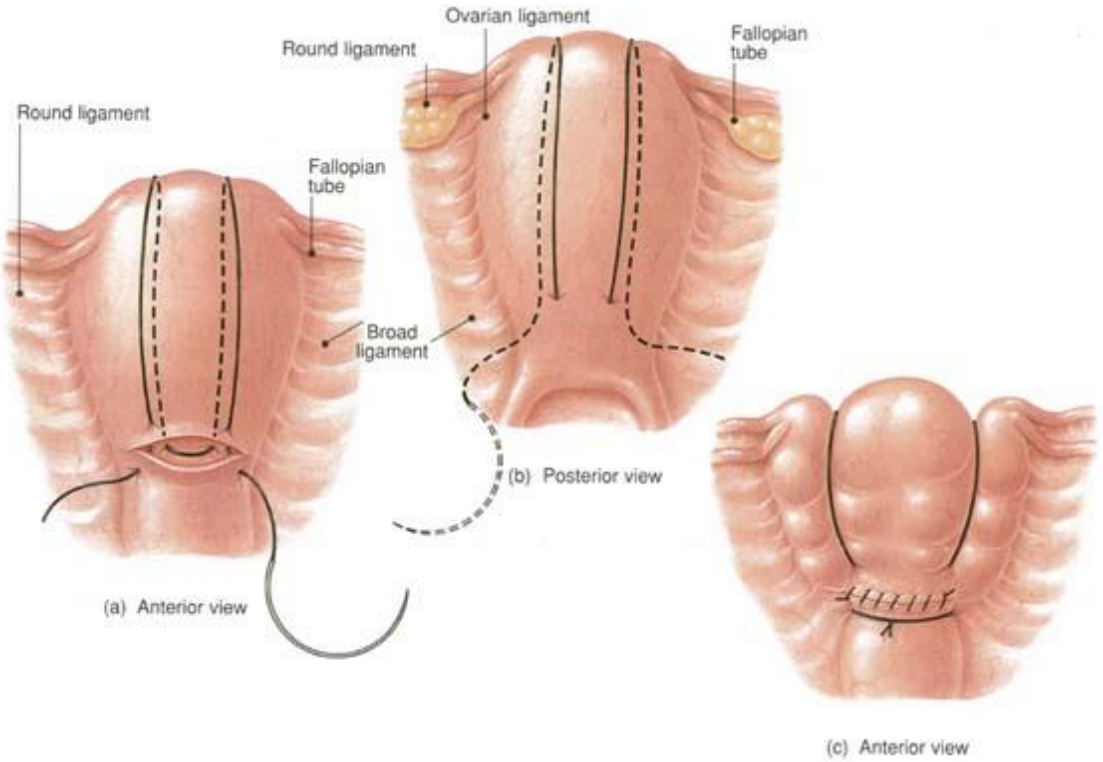
7.1 Intrauterine Balloon

- The woman is placed in lithotomy position.
- Insert Foley's urinary catheter, if not already present.
- Insert balloon catheter trans vaginally with sponge holding forceps or manually into uterine cavity ensuring that the entire balloon is inserted pass the cervical canal.
- Attach the enclosed giving set to the balloon using the 3 way tap.
- Fill either by rapid instillation with warm saline or use the 3 way tap and syringe for slow controlled filling.
- Review uterine tone and bleeding.
- Continue to insert saline until haemostasis has been achieved or until maximum volume **500** ml has been reached.
- Connect the drainage port to a fluid collection bag (catheter bag).
- Caution if placed after a B- Lynch suture has been used - because of the risk of uterine rupture.
- Document balloon placement on the avatar and apply a purple pack band to mum.
- Balloon should remain 12-24 hours.
- Cover with broad spectrum IV Antibiotics whilst it remains in situ.
- Following review – gradually deflate by 100mls an hour using bladder syringe.



7.2 B Lynch picture – Figure 1

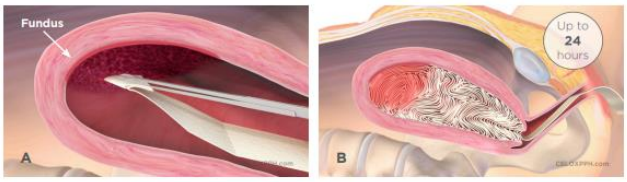
- N.o 1 Monocryl on 70mm blunt handheld needle
- Laminated picture available in theatre



7.3 Celox PPH

- The woman is placed in lithotomy position.
- Insert Foley's urinary catheter, if not already present.
- Use at least one Celox PPH for each patient and do not tear or cut prior to application
- Allow the Celox PPH to unfold externally and manually insert into the Uterine cavity – you could use a sponge holder.
- Ensure the Celox PPH is in contact with the bleeding site
- Following haemostasis excess Celox PPH can be cut from the end protruding from the vagina.
- Document pck placement on the avatar and apply a purple pack band to mum
- Celox PPH should remain 12-24 hours.
- Cover with broad spectrum IV Antibiotics whilst it remains in situ.

Figure 10. Application of CELOX™ PPH at site of Haemorrhage



Application of CELOX™ PPH at the fundus of the bleeding uterus (A). CELOX™ PPH is removed up to 24 hours after initial application (B), and after haemostasis has occured.

Indication for use

CELOX™ PPH is intended to be applied and removed vaginally to provide control and treatment of uterine postpartum haemorrhage when conservative management is warranted.

7.4 Emergency Radiological Intervention

Interventional radiology should be considered in the management of postpartum haemorrhage secondary to:

- atonic uterus following normal or prolonged labour, with or without caesarean section.
- surgical complications or uterine tears at the time of caesarean section.
- bleeding while on the recovery unit or in the postnatal ward following a normal delivery or a caesarean section.
- bleeding following hysterectomy.
- prophylactic treatment such as suspected case of placenta accretaie/placenta praevia on previous caesarean scar.

If required, it is performed in the radiology department at the request of the Consultant Obstetrician in consultation with the Consultant Anaesthetist.

Catheter balloons are inserted in the internal iliac or uterine arteries. Embolisation can then be performed via the balloon catheters if bleeding persistent and life threatening.

8.0 WOMEN WHO DECLINE BLOOD PRODUCTS

8.1 Where replacement blood products are required and the woman refuses to accept advice and treatment please refer to the Clinical guideline for women refusing blood or blood products in pregnancy, labour or postnatally.

9.0 SECONDARY PPH

9.1 Defined as bleeding after 24 hours post partum (commonest 8th - 14th day).

9.2 Causes

- Infection
- Retained products

9.3 Management

- High vaginal and endocervical swab
- Bloods - FBC, CRP, G&S, Coagulation studies
- Broad spectrum antibiotics (e.g. Augmentin or Cefalexin & Metronidazole see Antimicrobial Guidelines).
- Discuss with on call microbiologist if clinically septic and Urgent review by Obstetric Consultant
- Evacuation of uterus under anaesthetic if required – d/w Consultant Obstetrician
- As per the management of persistent bleeding following pregnancy flow chart (please add link) an ultrasound scan is not routinely recommended until after day 21.

10. PATIENT CARE FOLLOWING MAJOR HAEMORRHAGE

10.1 THE PLACE FOR RECOVERY OF THE PATIENT SHOULD BE DECIDED BY INDIVIDUAL CASE ON DISCUSSION WITH THE CONSULTANT OBSTETRICIAN AND ANAESTHETIST INVOLVED.

- ITU may be required if a massive transfusion has occurred.
- Otherwise all patients should have high dependency care within the labour ward environment.
- This will be guided by the anaesthetist on-call for labour ward and should include:
  - Continuous oxygen therapy.
  - Use of MEWS system.
  - 15 minute measurements of Respiratory Rate / Pulse / Blood Pressure with reduced frequency when stable.
  - Hourly urine output measurement aiming for $\geq$ 0.5 ml/kg/hour.
  - 4 hourly temperature / pain / sedation scores.
  - Repeat FBC / U&E / coagulation screen after 6 hours, then when required.
  - Frequent reassessment by the anaesthetic and obstetric medical team covering labour ward.
- The patient should be transferred back to the ward when all members of the team are happy she is fully recovered and stable.

11.0 DEBRIEFING THE STAFF AND FAMILY

11.1 Major obstetric haemorrhage can be traumatic to the woman, her family and the birth attendants; therefore, debriefing is recommended by a senior member of the team who was involved at the time of events at the earliest opportunity.

12.0 TRAINING

12.1 All birth attendants must attend annual ‘skills drills’ training on the management of post partum haemorrhage.

13.0 REFERENCES

B-Lynch C, Coker A, Adegboyega HC, Abu J, Cowen MJ. The B-Lynch surgical technique for the control of massive postpartum haemorrhages an alternative to hysterectomy? Five cases reported. *BJOG* 1997; **104:** 372-5.

RCOG Guideline – number 52 – Prevention and Management of Postpartum Haemorrhage - December 2016.

Saving Mother’s Lives: reviewing maternal deaths to make motherhood safer – 2003-2005.

The Seventh Report on Confidential Enquiries into Maternal Deaths in the United Kingdom. *London:CEMACH.*

Saving Live, Improving Mother’s Care – Lessons learned to inform maternity care from the UK and Ireland Confidential Enquiries into Maternal Deaths and Morbidity 2016-2018 (Dec 2020).

14.0 MONITORING COMPLIANCE WITH THIS GUIDELINE

14.1 Any concern or non-compliance with this guideline that is identified through the investigation of clinical incidents, claims or complaints will be reviewed as per the Trust Policies regarding Incidents, Claims and Complaints, and may result in an audit and/or amendment to the guideline.

14.2 Relevant Policies:

- Incident reporting policy and procedure
- Claims management policy and procedure
- Policy and Procedure for the Management of Complaints, Concerns, Comments and Compliments

15.0 ASSOCIATED CLINICAL GUIDELINES OR POLICIES

- Clinical guideline for care of women in labour in all care settings
- Clinical guideline for use of intraoperative cell salvage (ICS) in obstetrics
- Trust clinical guideline for massive blood loss
- Clinical guideline for maternal transfer by ambulance
- Clinical guideline for women refusing blood or blood products in pregnancy, labour or postnatally

16.0 PUBLICATION DETAILS

Author of Clinical Guideline	Labour Ward Lead Clinician
Division/ Department responsible for Guideline	Women's and Children's Health / CWH / Maternity services
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APPENDIX 1 - GUIDE TO THE MANAGEMENT OF OBSTETRIC HAEMORRHAGE

MAJOR OBSTETRIC HAEMORRHAGE

>1500ml blood loss

Give 1g Tranexamic Acid

STOP THE BLEEDING

Syntocinon 5iu iv + 5iu iv

40iu infusion over 3 hrs

Ergometrine 500mcg in 10ml, 1 ml at a time, surgeons to say if need more

Caution in PET

Carboprost 250mcg im

repeat every 15 mins

Caution in asthmatics

Misoprostol 800mcg PR/SL

Surgically

Bimanual compression

Balloon tamponade

B-Lynch suture

Hysterectomy

ACTIVATE 2222

I WANT TO ACTIVATE THE MOH

1 person to speak with lab

Emergency Group O immediately available if cant wait for xmatch

X2 Obs, x2 Main Theatre, x2 Lab

MOH Pack 1

X4 RBC Group Specific / X Match

REASSESS

MOH Pack 2

X4 RBC, x2 Cryprecipitate

If low fibrinogen or DIC likely Fibrinogen Concentrate from lab

RESUSCITATE A B C

Call for Help

x2 cannula

Take blood samples

Start TEG and Cell Salvage with Swab Wash

Think Cause!

Remember DIC

Antenatally:

placenta praevia/ abruption

Deliver immediately

Postnatally: 4Ts:

Tone/ Trauma/ Tissue/ Thrombin